

Chapter 1

Introduction

Commutative algebra is about studying commutative rings and their modules. The course has a view to algebraic geometry.

Example 1.1 (Examples of Commutative Rings). $\mathbb{Z}$, $k[x_1, \dots, x_n]$ for k field. For topological spaces, X we get the commutative ring $C(X) := \{f : X \rightarrow \mathbb{C} : f \text{ continuous}\}$.

Theorem 1.1. Structure Theorem for Finitely Generated Rings over a PID

Theorem 1.2. $k[x_1, \dots, x_n]$ is a UFD

Chapter 2

Basic Definitions

Definition 2.1 (Ring). A **ring** A is a triple $(A, +, \times)$ where A is a set and $+$ and $\times$ are binary operations on A .

1. $(A, +, 0)$ is an abelian group
2. $(A, 1, \times)$ is a monoid.
3. Distributive Laws: $(a + b) \times c = a \times c + b \times c$ and $a \times (b + c) = a \times b + a \times c$

A is a **commutative ring** if and only if $(A, 1, \times)$ is a commutative monoid.

Example 2.1. Examples of noncommutative rings are $M_n(k)$ where k is a field, kG for a nonabelian group G . Examples of commutative rings include $C^\infty(X)$ for X smooth manifold.

If a ring is discussed, then it is commutative from now on unless otherwise stated.

Note on the definition, $1 \in A$ (by definition), though this is not always required. A zero ring: $0 = 1 \in A$

Exercise 2.1. Show $\forall x \in A, 0 \cdot x = 0$ and $(-1)x = -x$

Definition 2.2 (Unit). An element $x \in A$ is called a unit (or invertible) if $\exists y \in A$ where $xy = 1 = yx$. An exercise is showing that y is unique. $A^\times$ is the group of units of A .

Definition 2.3 (Zero Divisor). An element $x \in A$ is a **zero divisor** if there exists a nonzero $y \in A$ where $xy = 0$.

Definition 2.4 (Nilpotent). $x \in A$ is **nilpotent** if $\exists n \in \mathbb{N}$ where $x^n = 0 \in A$

Definition 2.5 (Field). A is a field if $1 \neq 0$ and for every $x \neq 0$ in A is a unit.

Definition 2.6 (Integral Domain). A is an **integral domain** or **domain** if $1 \neq 0$ and $\forall x, y \in A, xy = 0 \Rightarrow x = 0$ or $y = 0$.

Definition 2.7 (Reduced). A ring A is **reduced** if only nilpotent element in A is 0.

We have the following inclusions:

$$\text{Field} \subset \text{Domain} \subset \text{Reduced}$$

Example 2.2. The zero ring is reduced, but not a domain (or a field). $\mathbb{Z}_p$ is a field if and only if p is prime in $\mathbb{Z}$. $\mathbb{Z}_n$ is not a field if and only if n is not prime. If $\mathbb{Z}_{p_1 \dots p_n}$ is reduced if $n \geq 0$ and $p_1, \dots, p_n$ are distinct.

Definition 2.8 (Subring). A subring of a ring R , $\{0, 1\} \subset A \subset R$ is a subset where A is closed under addition, additive inverses, multiplication, (in R)

Remark 2.1. $\{0\}$ is not a subring of $\mathbb{Z}$

Note in the category of commutative rings, $\mathbb{Z}$ is an initial object.

Definition 2.9 (Left and Right Ideal). An right ideal I in a ring A is a subset where $\forall k \in I$, then $kA \subseteq I$. A left ideal I is subset where $Ak \subset I$. In a commutative ring left and right ideals are equal. Kernel of a ring homomorphism is always a ring, furthermore any ideal is a kernel of a homomorphism

Example 2.3. In a field k , the only ideals are $\{0\}$ and k . If X is a topological space, $Y \subset X$ is a closed (if it isn't closed you can just take closure because of continuity) subspace. Let $I = \ker \left(C(X) \xrightarrow{|_Y} C(Y) \right)$ is a ideal. This is a typical way to get ideals, take a space of functions which is a ring and then restrict on a closed subspace.

Definition 2.10 (The Ideal Generated by a Set). The ideal generated by $S \subset A$ a ring denoted (S) :

1. The intersection of all ideals containing S .
2. The set of finite linear combinations of elements of S . If $S = \emptyset$. Thus set is the 0 ideal.

For a finite set $S = \{a_1, \dots, a_r\} \subset A$ we will used the notation: $(a_1, \dots, a_r) = (S)$.

Definition 2.11 (Principal Ideal). An ideal generated by one element.

Note, while the intersection of two ideals is always a ideal, the union is not always a ideal. But the sum of two ideals is a ideal, in fact the smallest ideal containing $I \cup J$.

Definition 2.12 (Quotient Ring). Standard way: R/I is the set of cosets but not ideal Better way: A/I is a ring that come with a ring homomorphism $\varphi : A \twoheadrightarrow A/I$ with kernel I .

Note for the sake of notation: For $a \in A$ we can say $a \in A/I$ to talk about $a + I \in A/I$.

Definition 2.13 (Maximal Ideals). A ideal I is called maximal if A/I is a field.

Definition 2.14 (Prime Ideals). A ideal I is a prime if A/I is a domain

Definition 2.15 (Radical Ideal). A ideal I is radical if A/I is reduced.

Example 2.4 (Maximal, Prime, and Radical Ideas). For a field k , the maximal ideal in $k[x]$ (which is also a PID) is (f) where f an monic irreducible polynomial.

Monic means leading coefficient is 1. In any **domain** R , $f \in R^\times$ and for $g, h \in R$ $f = gh \Rightarrow g$ or h is a unit for R .

Prime ideals of $k[x]$ are the maximal ideal and 0. Finally the radical ideals of $k[x]$ are (0) and $(f_1, \dots, f_r)$ for $1 \leq r < \infty$

Example 2.5 (Prime ideals in a UFD). $\mathbb{Z}[x]$ is a unique factorization domain and not a pid. Consider $(0), (7), (x), (7, x) \subset \mathbb{Z}[x]$. Only $(7, x)$ is maximal.

Example 2.6. Let S be a set, k a field, and A a ring of functions $S \rightarrow k$ that contains all constant function. Fix $s_0 \in S$

In “good cases”, in different points in S give different maximal ideals

Example 2.7. If X a compact Hausdorff space, or a metrizable topological space, by the Tietze Extension Theorem for any closed subset $Y \subset X$ for any continuous $f : Y \rightarrow \mathbb{R}$ there is a Extension $X \rightarrow \mathbb{R}$. Basically you can separate points hence the ideals generated are different.

Chapter 3

Spectrum of Ring

Definition 3.1 (Spectrum of a ring). For a commutative ring A ,

$$\text{Spec}(A) := \{\text{prime ideals in } A\}$$

Think of elements of A as functions on the space $\text{Spec}(A)$. So A turns into the ring of functions on $\text{Spec}(A)$. For each $\mathfrak{p} \in \text{Spec}(A)$, there is a homomorphism

$$A \rightarrow A/\mathfrak{p} \rightarrow \text{Frac}(A/\mathfrak{p}).$$

For a function $f \in A$, the value of f at $\mathfrak{p}$ is the image of f in this homomorphism.

Why are we taking the fraction field of a $A/\mathfrak{p}$?

3.1 Why only prime ideals?

We want to take values in a field k . $A \xrightarrow{\text{hom}} k$ maps A to a subring of k , which is a domain, hence the map has a prime ideal as a kernel.

Example 3.1 (Spectrum of Basic Rings).

1. $\text{Spec } 0 = \emptyset$
2. If $A \neq 0$, then $\text{Spec } A \neq \emptyset$ since A must have a maximal ideal hence a prime ideal.

3.2 Zariski Topology

We will define a topology on $\text{Spec}(A)$.

Definition 3.2 (Zariski Topology of $\text{Spec}(A)$). For an ideal $I \subseteq A$, denote

$$V(I) := \{p \in \text{Spec}(A) : I \subseteq p\}.$$

A subset of $\text{Spec}(A)$ is closed if it is equal to $V(I)$ for some $I \subseteq A$ ideal. A subset of $\text{Spec}(A)$ is open if it is the complement of $V(I)$ for some $I \subseteq A$ ideal.

Why does $V(I)$ consist of ideals containing I rather than being contained in I ?

Interpreting elements of A as functions on

$$V(I) := \{\text{pts where all functions in } I \text{ vanish}\}$$

3.2.1 Nonstandard Notion:

For elements $f_1, \dots, f_r \in A$ write $\{f_1 = 0, \dots, f_r = 0\} \subseteq \text{Spec}(A)$ for $V((f_1, \dots, f_r))$

Theorem 3.3. Show that the closed and open sets defined before form a Topology.

Example 3.2 (Some more Spec).

1. $\text{Spec}(\mathbb{Q}) = \{(0)\}$
2. $\text{Spec}(\mathbb{R}) = \{(0)\}$
3. $\text{Spec}(\mathbb{Z}) = \{(0)\} \cup \{(p) : p \text{ prime}\}.$

$$\{10 = 0\} = \{(2), (5)\}$$

Chapter 3.2.1

References